



2.1.2. Frequency Dependent Active Power Control

Many grid operators require a frequency depending active power control in order to ensure grid stability. The content of the FSM block from Figure 2 is shown in Figure 3.

To implement a frequency depending active power control mode first of all the frequency at the WPP PoC has to be measured (VAR(L+38)). The frequency signal stored in VAR(L+52) is taking into account measurement delays and dynamics of CON(J+64). Additionally the frequency is filtered with a first order lag element (CON(J+59)). To be able to investigate the wind farm response in case of under- and over frequency the frequency signal can be manipulated by defining an offset via VAR(L+59) to simulate a frequency event. This offset is added to the measured frequency and can be ramped. The ramp rate is defined by CON(J+69). By setting to zero the ramping of the frequency offset is deactivated. The P(f)-characteristic is defined by nine nodes (CON(J+38) to CON(55)) and can be related to different reference basis. ICON(I+10) and ICON(I+11) are defining the frequency control mode to be chosen for under- and over frequency.

The frequency signal used as control variable is stored in VAR(L+60).

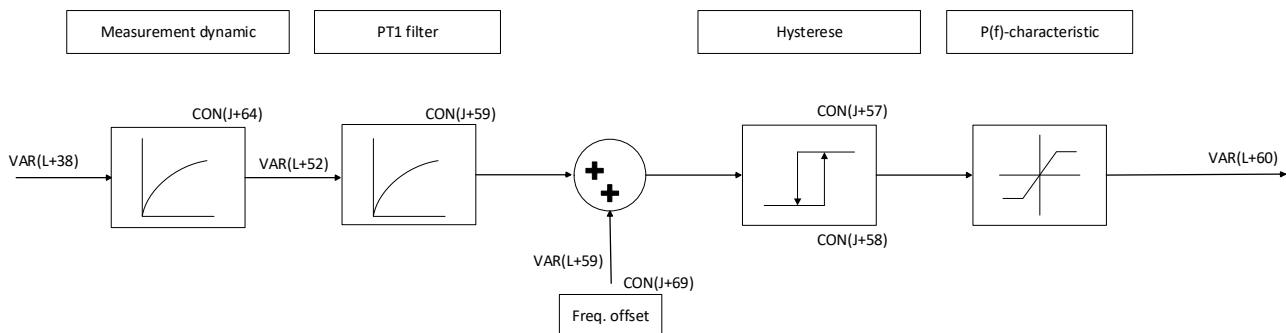


Figure 3: FSM block for P(f)-control within WFC

Following modes are available:

0: no control

1: Fixed active power roof

2: Floating active power roof / ramps in relation to rated power / starting from available power

3: Floating active power roof / ramps in relation to rated power / starting from instantaneous power

4: Floating active power roof / ramps in relation to instantaneous power / starting from instantaneous power

5: Floating active power roof / ramps in relation to available power / starting from available power

It has to be noted that the active power at model initialisation is used as available active power and fixed during the simulation.

Beside five different control modes also three hysteresis types are available for the floating active power roofs (mode: 2, 3, 4, 5) in case of overfrequency. ICON(I+12) is used to define the hysteresis mode: